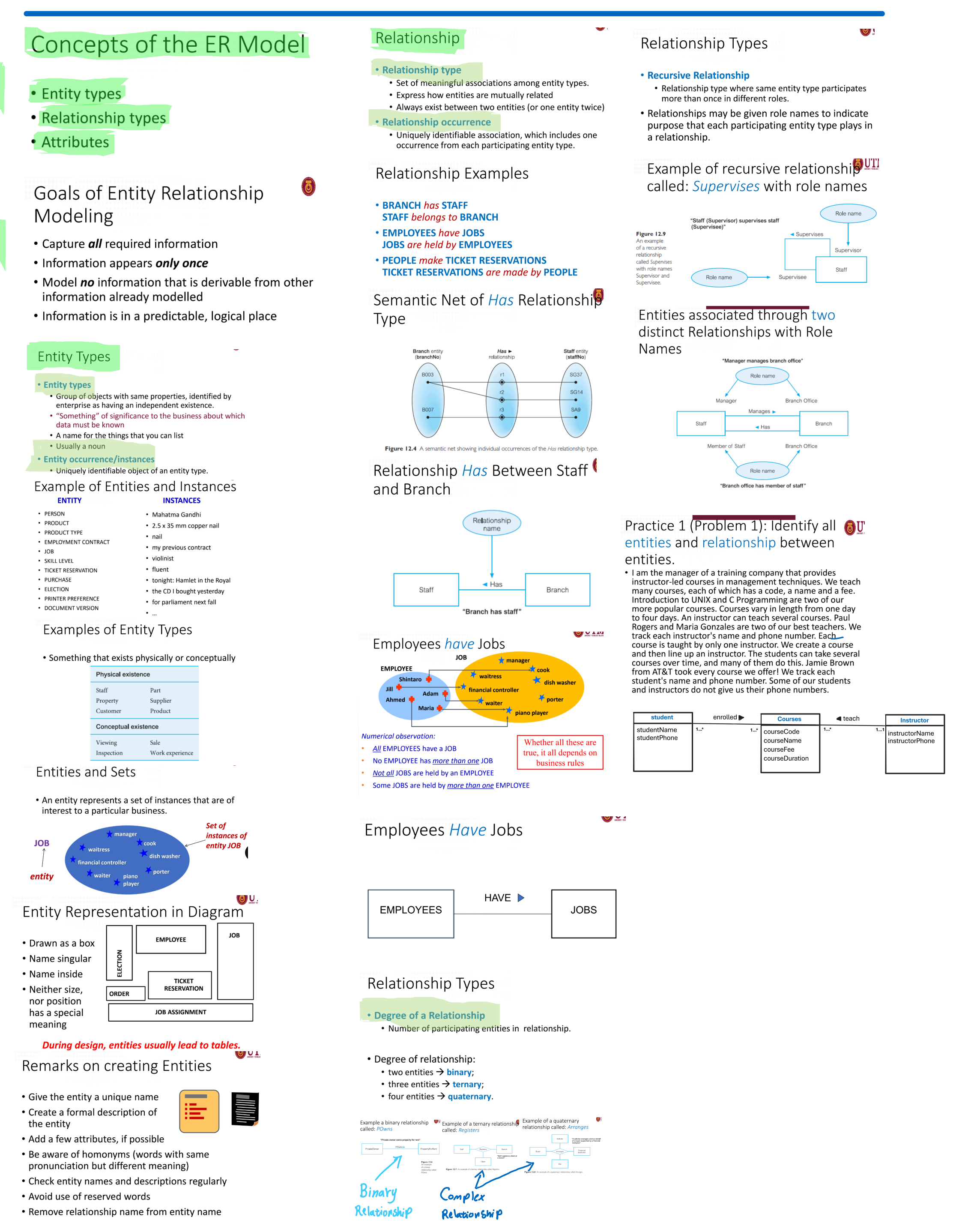
TOPIC 4 | ENTITY RELATIONSHIP MODELLING

- Models business, not implementation
- Is a well-established technique
- Has a robust syntax
- Results in easy-to-read diagrams…
- …although they may look rather complex at first sight

Part 1



Concepts of the ER Model

- Entity types
- Relationship types
- Attributes

Goals of Entity Relationship Modeling

- Capture ***all*** required information
- Information appears ***only once***
- Model ***no*** information that is derivable from other information already modelled
- Information is in a predictable, logical place

Entity Types

- Entity types
  - Group of objects with same properties, identified by enterprise as having an independent existence.
  - "Something" of significance to the business about which data must be known
  - A name for the things that you can list
  - Usually a noun
- Entity occurrence/instances
  - Uniquely identifiable object of an entity type.

Example of Entities and Instances

ENTITY	INSTANCES
PERSON	Mahatma Gandhi
PRODUCT	2.5 x 35 mm copper nail
PRODUCT TYPE	nail
EMPLOYMENT CONTRACT	my previous contract
JOB	violinist
SKILL LEVEL	fluent
TICKET RESERVATION	tonight: Hamlet in the Royal
PURCHASE	the CD I bought yesterday
ELECTION	for parliament next fall
PRINTER PREFERENCE	…
DOCUMENT VERSION	

Examples of Entity Types

- Something that exists physically or conceptually

Physical existence	
Staff	Part
Property	Supplier
Customer	Product
Conceptual existence	
Viewing	Sale
Inspection	Work experience

Entities and Sets

- An entity represents a set of instances that are of interest to a particular business.

JOB
entity
manager
cook
waitress
dish washer
financial controller
porter
waiter
piano player
Set of instances of entity JOB

Entity Representation in Diagram

- Drawn as a box
- Name singular
- Name inside
- Neither size, nor position has a special meaning

EMPLOYEE
JOB
ELECTION
TICKET RESERVATION
ORDER
JOB ASSIGNMENT

During design, entities usually lead to tables.

Remarks on creating Entities

- Give the entity a unique name
- Create a formal description of the entity
- Add a few attributes, if possible
- Be aware of homonyms (words with same pronunciation but different meaning)
- Check entity names and descriptions regularly
- Avoid use of reserved words
- Remove relationship name from entity name

Relationship

- Relationship type
  - Set of meaningful associations among entity types.
  - Express how entities are mutually related
  - Always exist between two entities (or one entity twice)
- Relationship occurrence
  - Uniquely identifiable association, which includes one occurrence from each participating entity type.

Relationship Examples

- **BRANCH** *has* **STAFF**
  STAFF *belongs to* **BRANCH**
- **EMPLOYEES** *have* **JOBS**
  JOBS *are held by* **EMPLOYEES**
- **PEOPLE** *make* **TICKET RESERVATIONS**
  TICKET RESERVATIONS *are made by* **PEOPLE**

Semantic Net of *Has* Relationship Type

Branch entity (branchNo)
Has relationship
Staff entity (staffNo)
B003
B007
r1
r2
r3
SG37
SG14
SA9

Figure 12.4 A semantic net showing individual occurrences of the *Has* relationship type.

Relationship *Has* Between Staff and Branch

Relationship name
Staff
Has
Branch

"Branch has staff"

Employees *have* Jobs

EMPLOYEE
JOB
Shintaro
Jill
Ahmed
Adam
Maria
manager
cook
waitress
dish washer
financial controller
porter
waiter
piano player

Numerical observation:

- *All* EMPLOYEES have a JOB
- No EMPLOYEE has *more than one* JOB
- *Not all* JOBS are held by an EMPLOYEE
- Some JOBS are held by *more than one* EMPLOYEE

Whether all these are true, it all depends on business rules

Employees *Have* Jobs

EMPLOYEES
HAVE
JOBS

Relationship Types

- Degree of a Relationship
  - Number of participating entities in relationship.
- Degree of relationship:
  - two entities → **binary**;
  - three entities → **ternary**;
  - four entities → **quaternary**.

Example of a binary relationship called: *POwns*

Example of a ternary relationship called: *Registers*

Example of a quaternary relationship called: *Arranges*

Binary Relationship

Complex Relationship

Relationship Types

- **Recursive Relationship**
  - Relationship type where same entity type participates more than once in different roles.
- Relationships may be given role names to indicate purpose that each participating entity type plays in a relationship.

Example of recursive relationship called: *Supervises* with role names

"Staff (Supervisor) supervises staff (Supervisee)"
Role name
Supervises
Supervisor
Staff
Supervisee
Role name

Figure 12.9 An example of a recursive relationship called Supervises with role names Supervisor and Supervisee.

Entities associated through two distinct Relationships with Role Names

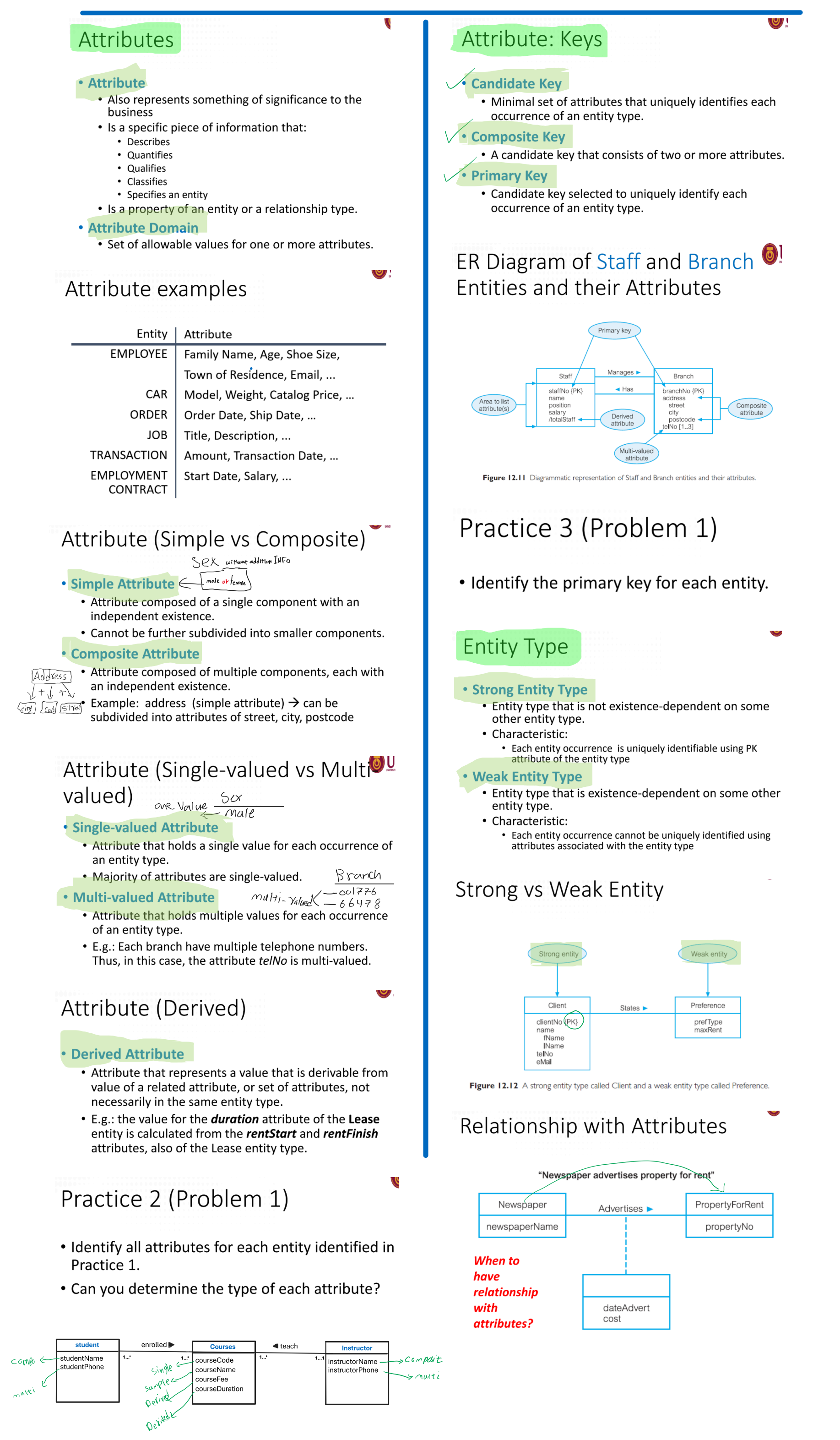
"Manager manages branch office"
Role name
Manager
Branch Office
Manages
Staff
Has
Branch
Member of Staff
Branch Office
Role name
"Branch office has member of staff"

Practice 1 (Problem 1): Identify all entities and relationship between entities.

- I am the manager of a training company that provides instructor-led courses in management techniques. We teach many courses, each of which has a code, a name and a fee. Introduction to UNIX and C Programming are two of our more popular courses. Courses vary in length from one day to four days. An instructor can teach several courses. Paul Rogers and Maria Gonzales are two of our best teachers. We track each instructor's name and phone number. Each course is taught by only one instructor. We create a course and then line up an instructor. The students can take several courses over time, and many of them do this. Jamie Brown from AT&T took every course we offer! We track each student's name and phone number. Some of our students and instructors do not give us their phone numbers.

student
studentName
studentPhone
enrolled
1…*
Courses
courseCode
courseName
courseFee
courseDuration
teach
1…1
Instructor
instructorName
instructorPhone

Part 2



Attributes

- Attribute
  - Also represents something of significance to the business
  - Is a specific piece of information that:
    - Describes
    - Quantifies
    - Qualifies
    - Classifies
    - Specifies an entity
  - Is a property of an entity or a relationship type.
- Attribute Domain
  - Set of allowable values for one or more attributes.

Attribute examples

Entity	Attribute
EMPLOYEE	Family Name, Age, Shoe Size, Town of Residence, Email, …
CAR	Model, Weight, Catalog Price, …
ORDER	Order Date, Ship Date, …
JOB	Title, Description, …
TRANSACTION	Amount, Transaction Date, …
EMPLOYMENT CONTRACT	Start Date, Salary, …

Attribute (Simple vs Composite)

Sex without addition INFO

male or female

- Simple Attribute
  - Attribute composed of a single component with an independent existence.
  - Cannot be further subdivided into smaller components.
- Composite Attribute
  - Attribute composed of multiple components, each with an independent existence.
  - Example: address (simple attribute) → can be subdivided into attributes of street, city, postcode

Address

city, code, Street

Attribute (Single-valued vs Multi-valued)

one Value Sex — male

- Single-valued Attribute
  - Attribute that holds a single value for each occurrence of an entity type.
  - Majority of attributes are single-valued.
- Multi-valued Attribute
  - Attribute that holds multiple values for each occurrence of an entity type.
  - E.g.: Each branch have multiple telephone numbers. Thus, in this case, the attribute *telNo* is multi-valued.

Branch
multi-Valued
001776
66478

Attribute (Derived)

- Derived Attribute
  - Attribute that represents a value that is derivable from value of a related attribute, or set of attributes, not necessarily in the same entity type.
  - E.g.: the value for the ***duration*** attribute of the **Lease** entity is calculated from the ***rentStart*** and ***rentFinish*** attributes, also of the Lease entity type.

Practice 2 (Problem 1)

- Identify all attributes for each entity identified in Practice 1.
- Can you determine the type of each attribute?

student
studentName
studentPhone
comp
multi
enrolled
Courses
courseCode
courseName
courseFee
courseDuration
single
sample
Derived
Derived
teach
Instructor
instructorName
instructorPhone
Composit
multi

Attribute: Keys

- Candidate Key
  - Minimal set of attributes that uniquely identifies each occurrence of an entity type.
- Composite Key
  - A candidate key that consists of two or more attributes.
- Primary Key
  - Candidate key selected to uniquely identify each occurrence of an entity type.

ER Diagram of Staff and Branch Entities and their Attributes

Primary key
Staff
staffNo {PK}
name
position
salary
/totalStaff
Manages
Has
Branch
branchNo {PK}
address
street
city
postcode
telNo [1..3]
Area to list attribute(s)
Derived attribute
Multi-valued attribute
Composite attribute

Figure 12.11 Diagrammatic representation of Staff and Branch entities and their attributes.

Practice 3 (Problem 1)

- Identify the primary key for each entity.

Entity Type

- Strong Entity Type
  - Entity type that is not existence-dependent on some other entity type.
  - Characteristic:
    - Each entity occurrence is uniquely identifiable using PK attribute of the entity type
- Weak Entity Type
  - Entity type that is existence-dependent on some other entity type.
  - Characteristic:
    - Each entity occurrence cannot be uniquely identified using attributes associated with the entity type

Strong vs Weak Entity

Strong entity
Weak entity
Client
clientNo {PK}
name
fName
lName
telNo
eMail
States
Preference
prefType
maxRent

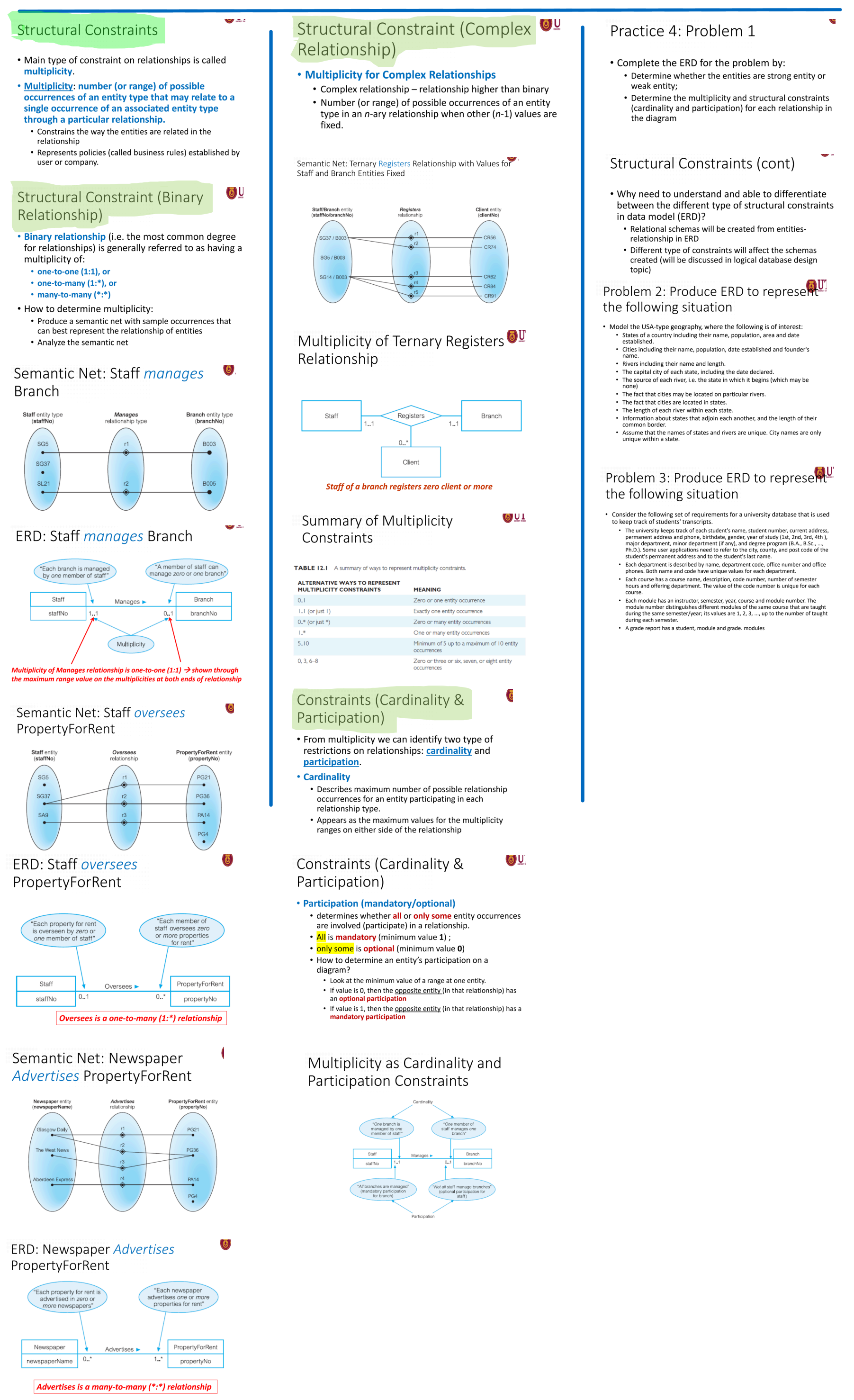
Figure 12.12 A strong entity type called Client and a weak entity type called Preference.

Relationship with Attributes

"Newspaper advertises property for rent"
Newspaper
newspaperName
Advertises
PropertyForRent
propertyNo
dateAdvert
cost

When to have relationship with attributes?

Part 3



Structural Constraints

- Main type of constraint on relationships is called **multiplicity**.
- **Multiplicity: number (or range) of possible occurrences of an entity type that may relate to a single occurrence of an associated entity type through a particular relationship.**
  - Constrains the way the entities are related in the relationship
  - Represents policies (called business rules) established by user or company.

Structural Constraint (Binary Relationship)

- **Binary relationship** (i.e. the most common degree for relationships) is generally referred to as having a multiplicity of:
  - **one-to-one (1:1), or**
  - **one-to-many (1:*), or**
  - **many-to-many (*:*)**
- How to determine multiplicity:
  - Produce a semantic net with sample occurrences that can best represent the relationship of entities
  - Analyze the semantic net

Semantic Net: Staff *manages* Branch

Staff entity type (staffNo)
Manages relationship type
Branch entity type (branchNo)
SG5
SG37
SL21
r1
r2
B003
B005

ERD: Staff *manages* Branch

"Each branch is managed by one member of staff"
"A member of staff can manage zero or one branch"
Staff
staffNo
Manages
1..1
0..1
Branch
branchNo
Multiplicity

Multiplicity of Manages relationship is one-to-one (1:1) → shown through the maximum range value on the multiplicities at both ends of relationship

Semantic Net: Staff *oversees* PropertyForRent

Staff entity (staffNo)
Oversees relationship
PropertyForRent entity (propertyNo)
SG5
SG37
SA9
r1
r2
r3
PG21
PG36
PA14
PG4

ERD: Staff *oversees* PropertyForRent

"Each property for rent is overseen by zero or one member of staff"
"Each member of staff oversees zero or more properties for rent"
Staff
staffNo
Oversees
0..1
0..*
PropertyForRent
propertyNo

Oversees is a one-to-many (1:) relationship*

Semantic Net: Newspaper *Advertises* PropertyForRent

Newspaper entity (newspaperName)
Advertises relationship
PropertyForRent entity (propertyNo)
Glasgow Daily
The West News
Aberdeen Express
r1
r2
r3
r4
PG21
PG36
PA14
PG4

ERD: Newspaper *Advertises* PropertyForRent

"Each property for rent is advertised in zero or more newspapers"
"Each newspaper advertises one or more properties for rent"
Newspaper
newspaperName
Advertises
0..*
1..*
PropertyForRent
propertyNo

Advertises is a many-to-many (:*) relationship*

Structural Constraint (Complex Relationship)

- **Multiplicity for Complex Relationships**
  - Complex relationship – relationship higher than binary
  - Number (or range) of possible occurrences of an entity type in an *n*-ary relationship when other (*n*-1) values are fixed.

Semantic Net: Ternary Registers Relationship with Values for Staff and Branch Entities Fixed

Staff/Branch entity (staffNo/branchNo)
Registers relationship
Client entity (clientNo)
SG37 / B003
SG5 / B003
SG14 / B003
r1
r2
r3
r4
r5
CR56
CR74
CR62
CR84
CR91

Multiplicity of Ternary Registers Relationship

Staff
Registers
Branch
1..1
1..1
0..*
Client

Staff of a branch registers zero client or more

Summary of Multiplicity Constraints

TABLE 12.1 A summary of ways to represent multiplicity constraints.

ALTERNATIVE WAYS TO REPRESENT MULTIPLICITY CONSTRAINTS	MEANING
0..1	Zero or one entity occurrence
1..1 (or just 1)	Exactly one entity occurrence
0..* (or just *)	Zero or many entity occurrences
1..*	One or many entity occurrences
5..10	Minimum of 5 up to a maximum of 10 entity occurrences
0, 3, 6–8	Zero or three or six, seven, or eight entity occurrences

Constraints (Cardinality & Participation)

- From multiplicity we can identify two type of restrictions on relationships: cardinality and participation.
- Cardinality
  - Describes maximum number of possible relationship occurrences for an entity participating in each relationship type.
  - Appears as the maximum values for the multiplicity ranges on either side of the relationship

Constraints (Cardinality & Participation)

- Participation (mandatory/optional)
  - determines whether **all** or **only some** entity occurrences are involved (participate) in a relationship.
  - **All** is **mandatory** (minimum value **1**) ;
  - **only some** is **optional** (minimum value **0**)
  - How to determine an entity's participation on a diagram?
    - Look at the minimum value of a range at one entity.
    - If value is 0, then the opposite entity (in that relationship) has an **optional participation**
    - If value is 1, then the opposite entity (in that relationship) has a **mandatory participation**

Multiplicity as Cardinality and Participation Constraints

Cardinality
"One branch is managed by one member of staff"
"One member of staff manages one branch"
Staff
staffNo
Manages
1..1
0..1
Branch
branchNo
"All branches are managed" (mandatory participation for branch)
"Not all staff manage branches" (optional participation for staff)
Participation

Practice 4: Problem 1

- Complete the ERD for the problem by:
  - Determine whether the entities are strong entity or weak entity;
  - Determine the multiplicity and structural constraints (cardinality and participation) for each relationship in the diagram

Structural Constraints (cont)

- Why need to understand and able to differentiate between the different type of structural constraints in data model (ERD)?
  - Relational schemas will be created from entities-relationship in ERD
  - Different type of constraints will affect the schemas created (will be discussed in logical database design topic)

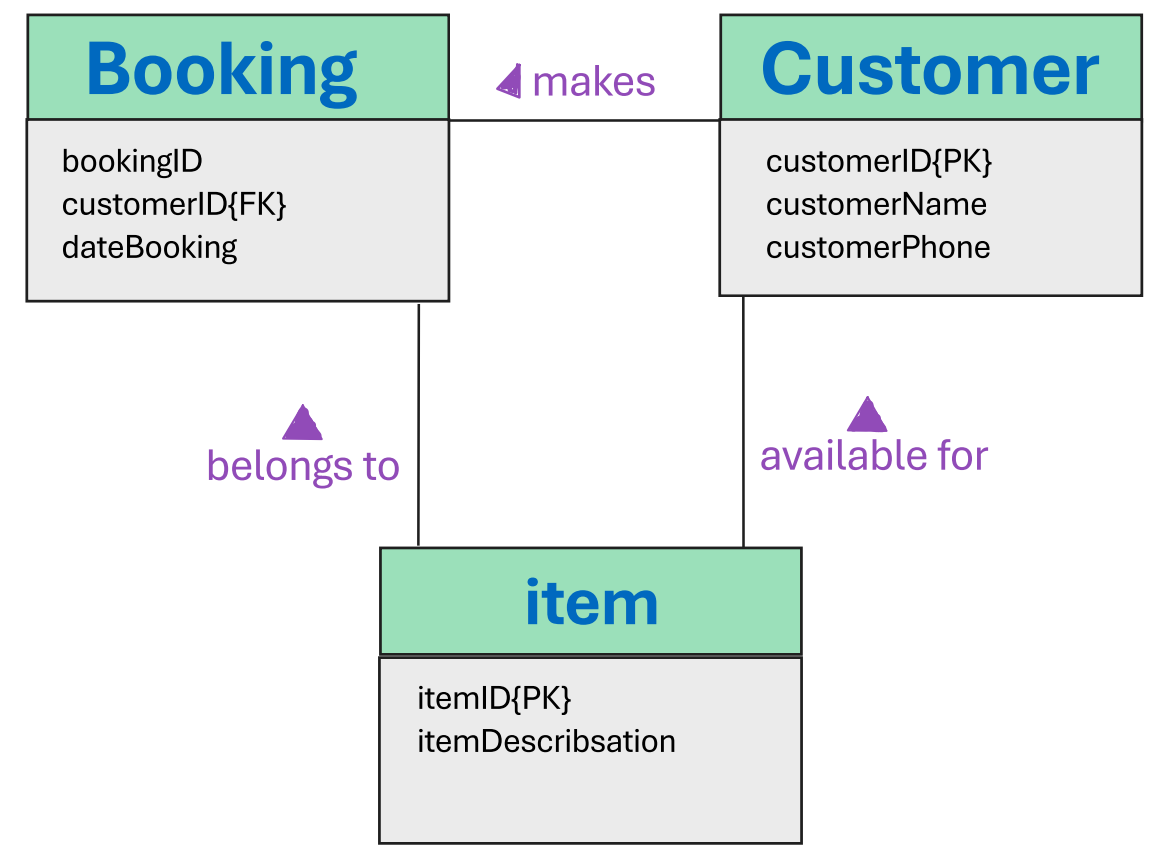
Problem 2: Produce ERD to represent the following situation

- Model the USA-type geography, where the following is of interest:
  - States of a country including their name, population, area and date established.
  - Cities including their name, population, date established and founder's name.
  - Rivers including their name and length.
  - The capital city of each state, including the date declared.
  - The source of each river, i.e. the state in which it begins (which may be none)
  - The fact that cities may be located on particular rivers.
  - The fact that cities are located in states.
  - The length of each river within each state.
  - Information about states that adjoin each another, and the length of their common border.
  - Assume that the names of states and rivers are unique. City names are only unique within a state.

Problem 3: Produce ERD to represent the following situation

- Consider the following set of requirements for a university database that is used to keep track of students' transcripts.
  - The university keeps track of each student's name, student number, current address, permanent address and phone, birthdate, gender, year of study (1st, 2nd, 3rd, 4th), major department, minor department (if any), and degree program (B.A., B.Sc., …, Ph.D.). Some user applications need to refer to the city, county, and post code of the student's permanent address and to the student's last name.
  - Each department is described by name, department code, office number and office phones. Both name and code have unique values for each department.
  - Each course has a course name, description, code number, number of semester hours and offering department. The value of the code number is unique for each course.
  - Each module has an instructor, semester, year, course and module number. The module number distinguishes different modules of the same course that are taught during the same semester/year; its values are 1, 2, 3, …, up to the number of taught during each semester.
  - A grade report has a student, module and grade. modules

15. Using the given scenario of a rental system (Rental Items, Customers, and Bookings), draw an ER diagram showing the entities and relationships. Include attributes for each entity. Indicate the multiplicity of relationships.


Booking
bookingID
customerID(FK)
dateBooking
makes
Customer
customerID(PK)
customerName
customerPhone
belongs to
available for
item
itemID(PK)
itemDescribsation